EXPERIMENT 7

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Build a deep recurrent neural network (RNN) to generate image captions. Combine CNN for image feature extraction with RNN for sequence generation.

```
import tensorflow as tf
import numpy as np
from tensorflow.keras import layers, Model

# Load CIFAR-10
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()

# Use small data for phone
x_train = x_train[:2000] / 255.0
y_train = y_train[:2000].flatten()

# CIFAR-10 class names
classes = [
    "airplane", "automobile", "bird", "cat", "deer",
    "dog", "frog", "horse", "ship", "truck"
]

# CNN for image feature extraction
cnn = tf.keras.applications.MobileNetV2(
    input_shape=(32, 32, 3),
    include_top=False,
    weights=None
)

cnn.trainable = False

# Resize images for MobileNet
images = tf.image.resize(x_train, (32, 32))

# Extract image features
features = cnn.predict(images, batch_size=32, verbose=0)

print("CNN Feature Shape:", features.shape)

# Convert features into a sequence
features = features.reshape(features.shape[0], -1, features.shape[-1])

# Caption words
vocab = [
    "a", "photo", "of", "an", "airplane",
    "automobile", "bird", "cat", "deer",
    "dog", "frog", "horse", "ship", "truck"
]

word_to_id = {word: i + 1 for i, word in enumerate(vocab)}

# Create simple captions
captions = []

for label in y_train:
    word = classes[label]
    captions.append([
        word_to_id["a"],
        word_to_id["photo"],
        word_to_id["of"],
        word_to_id[word]
    ])

captions = np.array(captions)

# RNN model
input_image = layers.Input(shape=features.shape[1:])
x = layers.LSTM(64)(input_image)
x = layers.Dense(32, activation="relu")(x)
output = layers.Dense(len(vocab) + 1, activation="softmax")(x)

model = Model(input_image, output)

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
# Train using last word as target
target = captions[:, -1]

print("Training RNN...")
model.fit(
    features,
    target,
    epochs=3,
    batch_size=32
)

# Generate caption
index = 0

prediction = model.predict(
    features[index:index+1],
    verbose=0
)

predicted_id = np.argmax(prediction)

predicted_word = vocab[predicted_id - 1] \
    if predicted_id > 0 else "unknown"

print("\nGenerated Caption:")
print("A photo of", predicted_word)
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 1386s 8us/step
CNN Feature Shape: (2000, 1, 1, 1280)
Training RNN...
Epoch 1/3
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.0975 - loss: 2.6501
Epoch 2/3
63/63 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.1015 - loss: 2.4461
Epoch 3/3
63/63 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.0990 - loss: 2.3357

Generated Caption:
A photo of deer
```